

Attention Is All You Need: The Transformer Architecture

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ABSTRACT

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 41.8 BLEU on the WMT 2014 English-to-French translation task.

1. INTRODUCTION

Recurrent neural networks, long short-term memory (LSTM) and gated recurrent neural networks have been firmly established as state of the art approaches in sequence modeling. In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output.

2. PROPOSED METHOD

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. Multi-Head Attention allows the model to jointly attend to information from different representation subspaces at different positions.

3. EXPERIMENTAL SETUP

We trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million sentence pairs. Sentences were encoded using byte-pair encoding, which has a shared source-target vocabulary of about 37000 tokens.

4. EXPERIMENTAL RESULTS

On the translation benchmarks, the Transformer outperforms existing architectures: Transformer-Big: 41.8 BLEU. Transformer-Base: 38.4 BLEU. ConvS2S: 32.8 BLEU. ByteNet: 23.7 BLEU. GNMT: 24.6 BLEU. Training cost was reduced by over 80% compared to recurrent baselines.

5. LIMITATIONS

Self-attention computes pairwise token similarities with $O(N^2)$ complexity with respect to sequence length, creating memory bottlenecks on very long documents.

6. FUTURE WORK

Future research will explore sparse attention patterns, linear attention approximations, and multi-modal audio-visual applications.

7. CONCLUSION

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention.